

Name: Adarsh Chandel

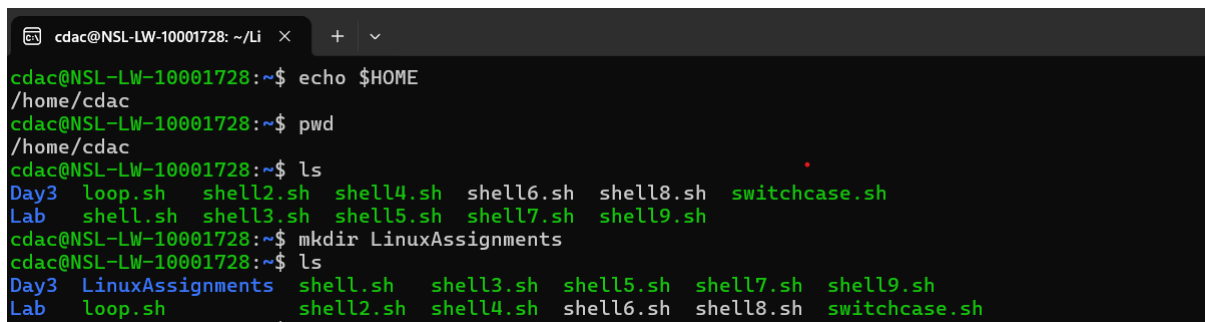
Concepts of Operating Systems

Assignment 1

Problem 1: Read the instructions carefully and answer accordingly. If there is any need to insert some data then do that as well.

a) Navigate and List:

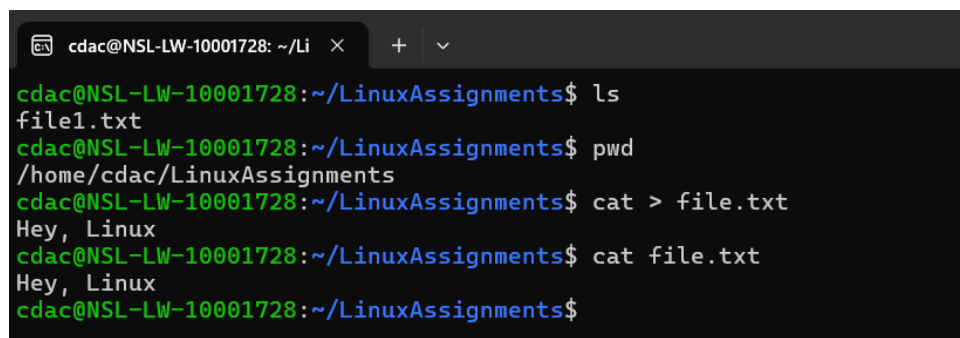
a. Start by navigating to your home directory and list its contents. Then, move into a directory named "LinuxAssignment" if it exists; otherwise, create it.



```
cdac@NSL-LW-10001728: ~/Li x + v
cdac@NSL-LW-10001728:~$ echo $HOME
/home/cdac
cdac@NSL-LW-10001728:~$ pwd
/home/cdac
cdac@NSL-LW-10001728:~$ ls
Day3  loop.sh  shell2.sh  shell4.sh  shell6.sh  shell8.sh  switchcase.sh
Lab   shell.sh  shell3.sh  shell5.sh  shell7.sh  shell9.sh
cdac@NSL-LW-10001728:~$ mkdir LinuxAssignments
cdac@NSL-LW-10001728:~$ ls
Day3  LinuxAssignments  shell.sh  shell3.sh  shell5.sh  shell7.sh  shell9.sh
Lab   loop.sh           shell2.sh  shell4.sh  shell6.sh  shell8.sh  switchcase.sh
```

b) File Management:

a. Inside the "LinuxAssignment" directory, create a new file named "file1.txt". Display its contents.



```
cdac@NSL-LW-10001728: ~/Li x + v
cdac@NSL-LW-10001728:~/LinuxAssignments$ ls
file1.txt
cdac@NSL-LW-10001728:~/LinuxAssignments$ pwd
/home/cdac/LinuxAssignments
cdac@NSL-LW-10001728:~/LinuxAssignments$ cat > file.txt
Hey, Linux
cdac@NSL-LW-10001728:~/LinuxAssignments$ cat file.txt
Hey, Linux
cdac@NSL-LW-10001728:~/LinuxAssignments$
```

c) Directory Management:

a. Create a new directory named "docs" inside the "LinuxAssignment" directory.

```
cdac@NSL-LW-10001728: ~/Li × + v
cdac@NSL-LW-10001728:~/LinuxAssignments$ pwd
/home/cdac/LinuxAssignments
cdac@NSL-LW-10001728:~/LinuxAssignments$ mkdir docs
cdac@NSL-LW-10001728:~/LinuxAssignments$ ls
docs  file.txt  file1.txt
cdac@NSL-LW-10001728:~/LinuxAssignments$ |
```

d) Copy and Move Files:

- a. Copy the "file1.txt" file into the "docs" directory and rename it to "file2.txt".

```
cdac@NSL-LW-10001728: ~/Li × + v
cdac@NSL-LW-10001728:~/LinuxAssignments$ ls
docs  file1.txt
cdac@NSL-LW-10001728:~/LinuxAssignments$ cp file1.txt docs/file2.txt
cdac@NSL-LW-10001728:~/LinuxAssignments$ cd docs
cdac@NSL-LW-10001728:~/LinuxAssignments/docs$ ls
file2.txt
cdac@NSL-LW-10001728:~/LinuxAssignments/docs$ cat file2.txt
Hey, Linux!!
```

e) Permissions and Ownership:

- a. Change the permissions of "file2.txt" to allow read, write, and execute permissions for the owner and only read permissions for others. Then, change the owner of "file2.txt" to the current user.

```
cdac@NSL-LW-10001728: ~/Li × + v
cdac@NSL-LW-10001728:~/LinuxAssignments/docs$ pwd
/home/cdac/LinuxAssignments/docs
cdac@NSL-LW-10001728:~/LinuxAssignments/docs$ ls -l
total 4
-rw-r--r-- 1 cdac cdac 28 Aug 21 11:48 file2.txt
cdac@NSL-LW-10001728:~/LinuxAssignments/docs$ chmod 744 file2.txt
cdac@NSL-LW-10001728:~/LinuxAssignments/docs$ ls -l
total 4
-rwxr--r-- 1 cdac cdac 28 Aug 21 11:48 file2.txt
cdac@NSL-LW-10001728:~/LinuxAssignments/docs$ |
```

```
cdac@NSL-LW-10001728: ~/Li × + v
cdac@NSL-LW-10001728:~/LinuxAssignments/docs$ whoami
cdac
cdac@NSL-LW-10001728:~/LinuxAssignments/docs$ chown cdac file2.txt
cdac@NSL-LW-10001728:~/LinuxAssignments/docs$ ls -l
total 4
-rwxr--r-- 1 cdac cdac 28 Aug 21 11:48 file2.txt
cdac@NSL-LW-10001728:~/LinuxAssignments/docs$
```

f) Final Checklist:

a. Finally, list the contents of the "LinuxAssignment" directory and the root directory to ensure that all operations were performed correctly.

```
root@NSL-LW-10001728: / × + v
cdac@NSL-LW-10001728:~/LinuxAssignments/docs$ pwd
/home/cdac/LinuxAssignments/docs
cdac@NSL-LW-10001728:~/LinuxAssignments/docs$ cd ..
cdac@NSL-LW-10001728:~/LinuxAssignments$ ls -l
total 8
drwxr-xr-x 2 cdac cdac 4096 Aug 21 11:48 docs
-rw-r--r-- 1 cdac cdac 28 Aug 21 11:42 file1.txt
```

```
/root/.hushlogin file.
root@NSL-LW-10001728:~# cd /
root@NSL-LW-10001728:/# ls -l
total 2740
lrwxrwxrwx 1 root root 7 Apr 22 2024 bin -> usr/bin
drwxr-xr-x 2 root root 4096 Feb 26 2024 bin.usr-is-merged
drwxr-xr-x 2 root root 4096 Apr 22 2024 boot
drwxr-xr-x 15 root root 3860 Aug 21 11:04 dev
drwxr-xr-x 88 root root 4096 Aug 21 11:04 etc
drwxr-xr-x 3 root root 4096 Aug 19 11:35 home
-rwxrwxrwx 1 root root 2724480 Jul 31 14:56 init
lrwxrwxrwx 1 root root 7 Apr 22 2024 lib -> usr/lib
drwxr-xr-x 2 root root 4096 Apr 8 2024 lib.usr-is-merged
lrwxrwxrwx 1 root root 9 Apr 22 2024 lib64 -> usr/lib64
drwx----- 2 root root 16384 Aug 19 11:34 lost+found
drwxr-xr-x 2 root root 4096 Aug 5 16:55 media
drwxr-xr-x 6 root root 4096 Aug 19 11:35 mnt
drwxr-xr-x 2 root root 4096 Aug 5 16:55 opt
dr-xr-xr-x 281 root root 0 Aug 21 11:04 proc
drwx----- 3 root root 4096 Aug 21 11:59 root
drwxr-xr-x 19 root root 580 Aug 21 11:59 run
lrwxrwxrwx 1 root root 8 Apr 22 2024 sbin -> usr/sbin
drwxr-xr-x 2 root root 4096 Mar 31 2024 sbin.usr-is-merged
drwxr-xr-x 2 root root 4096 Aug 19 11:35 snap
drwxr-xr-x 2 root root 4096 Aug 5 16:55 srv
dr-xr-xr-x 13 root root 0 Aug 21 11:04 sys
drwxrwxrwt 8 root root 4096 Aug 21 11:30 tmp
drwxr-xr-x 12 root root 4096 Aug 5 16:55 usr
drwxr-xr-x 13 root root 4096 Aug 19 11:35 var
root@NSL-LW-10001728:/#
```

g) File Searching:

- a. Search for all files with the extension ".txt" in the current directory and its subdirectories.
- b. Display lines containing a specific word in a file (provide a file name and the specific word to search).

```
cdac@NSL-LW-10001728: ~/Li × + v
cdac@NSL-LW-10001728:~$ ls
Day3 LinuxAssignments shell.sh shell3.sh shell5.sh shell7.sh shell9.sh
Lab loop.sh shell2.sh shell4.sh shell6.sh shell8.sh switchcase.sh
cdac@NSL-LW-10001728:~$ grep -r ".txt" /LinuxAssignments
grep: /LinuxAssignments: No such file or directory
cdac@NSL-LW-10001728:~$ grep -r ".txt" LinuxAssignments/
cdac@NSL-LW-10001728:~$ ls -R /home/cdac/LinuxAssignments | grep '\.txt$'
file1.txt
file2.txt
cdac@NSL-LW-10001728:~$ cd LinuxAssignments/
cdac@NSL-LW-10001728:~/LinuxAssignments$ cat > file3.txt
This is a test. This test is only a test.
In case of an actual emergency, the word test would appear more frequently.
Test the test by testing the test cases.
Repetition is key in a test. Test again. Test again.
cdac@NSL-LW-10001728:~/LinuxAssignments$ cat file3.txt
This is a test. This test is only a test.
In case of an actual emergency, the word test would appear more frequently.
Test the test by testing the test cases.
Repetition is key in a test. Test again. Test again.
cdac@NSL-LW-10001728:~/LinuxAssignments$ grep -i "test" file3.txt
This is a test. This test is only a test.
In case of an actual emergency, the word test would appear more frequently.
Test the test by testing the test cases.
Repetition is key in a test. Test again. Test again.
cdac@NSL-LW-10001728:~/LinuxAssignments$ |
```

h) System Information:

- a. Display the current system date and time.

```
cdac@NSL-LW-10001728: ~/Li × + v
cdac@NSL-LW-10001728:~/LinuxAssignments$ date
Thu Aug 21 12:15:40 UTC 2025
cdac@NSL-LW-10001728:~/LinuxAssignments$ |
```

i) Networking

a. Display the IP address of the system.

b. Ping a remote server to check connectivity (provide a remote server address to ping).

```
cdac@NSL-LW-10001728: ~  
cdac@NSL-LW-10001728:~$ ip a  
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000  
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00  
    inet 127.0.0.1/8 scope host lo  
        valid_lft forever preferred_lft forever  
    inet 10.255.255.254/32 brd 10.255.255.254 scope global lo  
        valid_lft forever preferred_lft forever  
    inet6 ::1/128 scope host  
        valid_lft forever preferred_lft forever  
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP group default qlen 1000  
    link/ether 00:15:5d:da:6f:e4 brd ff:ff:ff:ff:ff:ff  
    inet 172.18.242.181/20 brd 172.18.255.255 scope global eth0  
        valid_lft forever preferred_lft forever  
    inet6 fe80::215:5dff:feda:6fe4/64 scope link  
        valid_lft forever preferred_lft forever  
cdac@NSL-LW-10001728:~$ |
```

```
valid_lft forever preferred_lft forever  
cdac@NSL-LW-10001728:~$ ping github.com  
PING github.com (20.207.73.82) 56(84) bytes of data.  
64 bytes from 20.207.73.82: icmp_seq=1 ttl=111 time=95.6 ms  
64 bytes from 20.207.73.82: icmp_seq=2 ttl=111 time=43.0 ms  
64 bytes from 20.207.73.82: icmp_seq=3 ttl=111 time=41.0 ms  
64 bytes from 20.207.73.82: icmp_seq=4 ttl=111 time=43.8 ms  
64 bytes from 20.207.73.82: icmp_seq=5 ttl=111 time=42.6 ms  
64 bytes from 20.207.73.82: icmp_seq=6 ttl=111 time=40.5 ms  
64 bytes from 20.207.73.82: icmp_seq=7 ttl=111 time=43.3 ms  
64 bytes from 20.207.73.82: icmp_seq=8 ttl=111 time=42.1 ms  
64 bytes from 20.207.73.82: icmp_seq=9 ttl=111 time=41.6 ms  
64 bytes from 20.207.73.82: icmp_seq=10 ttl=111 time=44.2 ms  
64 bytes from 20.207.73.82: icmp_seq=11 ttl=111 time=43.6 ms  
64 bytes from 20.207.73.82: icmp_seq=12 ttl=111 time=43.0 ms  
^C  
--- github.com ping statistics ---  
12 packets transmitted, 12 received, 0% packet loss, time 11012ms  
rtt min/avg/max/mdev = 40.521/47.017/95.607/14.690 ms  
cdac@NSL-LW-10001728:~$ |
```

j) File Compression:

- a. Compress the "docs" directory into a zip file.
- b. Extract the contents of the zip file into a new directory.

```
cdac@NSL-LW-10001728: ~  
cdac@NSL-LW-10001728:~$ ls  
Day3  LinuxAssignments  shell.sh  shell3.sh  shell5.sh  shell7.sh  shell9.sh  
Lab   loop.sh             shell2.sh shell4.sh  shell6.sh  shell8.sh  switchcase.sh  
cdac@NSL-LW-10001728:~$ pwd  
/home/cdac  
cdac@NSL-LW-10001728:~$ cd LinuxAssignments  
cdac@NSL-LW-10001728:~/LinuxAssignments$ zip -r docs  
  
zip error: Nothing to do! (docs.zip)  
cdac@NSL-LW-10001728:~/LinuxAssignments$ zip -r docs.zip docs  
  adding: docs/ (stored 0%)  
  adding: docs/file2.txt (stored 0%)  
cdac@NSL-LW-10001728:~/LinuxAssignments$ ls  
docs  docs.zip  file1.txt  file3.txt
```

```
cdac@NSL-LW-10001728:~/LinuxAssignments$ ls  
docs  docs.zip  file1.txt  file3.txt  
cdac@NSL-LW-10001728:~/LinuxAssignments$ cd ..  
cdac@NSL-LW-10001728:~$ ls  
Day3  LinuxAssignments  shell.sh  shell3.sh  shell5.sh  shell7.sh  shell9.sh  
Lab   loop.sh             shell2.sh shell4.sh  shell6.sh  shell8.sh  switchcase.sh  
cdac@NSL-LW-10001728:~$ unzip docs.zip  
unzip: cannot find or open docs.zip, docs.zip.zip or docs.zip.ZIP.  
cdac@NSL-LW-10001728:~$ cd LinuxAssignments/  
cdac@NSL-LW-10001728:~/LinuxAssignments$ ls  
docs  docs.zip  file1.txt  file3.txt  
cdac@NSL-LW-10001728:~/LinuxAssignments$ cd ..  
cdac@NSL-LW-10001728:~$ cd ..  
cdac@NSL-LW-10001728:/home$ cd cdac  
cdac@NSL-LW-10001728:~$ mkdir unzipfile  
cdac@NSL-LW-10001728:~$ cd unzipfile/  
cdac@NSL-LW-10001728:~/unzipfile$ cd /home/cdac/LinuxAssignments/  
cdac@NSL-LW-10001728:~/LinuxAssignments$ ls  
docs  docs.zip  file1.txt  file3.txt  
cdac@NSL-LW-10001728:~/LinuxAssignments$ cp docs.zip /home/cdac/unzipfile/  
cdac@NSL-LW-10001728:~/LinuxAssignments$ cd /home/cdac/unzipfile/  
cdac@NSL-LW-10001728:~/unzipfile$ ls  
docs.zip  
cdac@NSL-LW-10001728:~/unzipfile$ unzip docs.zip  
Archive:  docs.zip  
  creating: docs/  
  extracting: docs/file2.txt  
cdac@NSL-LW-10001728:~/unzipfile$ ls  
docs  docs.zip  
cdac@NSL-LW-10001728:~/unzipfile$ cd docs  
cdac@NSL-LW-10001728:~/unzipfile/docs$ ls  
file2.txt  
cdac@NSL-LW-10001728:~/unzipfile/docs$ cat file2.txt  
Hey, Linux!!  
  
Good Morning
```

k) File Editing:

- a. Open the "file1.txt" file in a text editor and add some text to it.
- b. Replace a specific word in the "file1.txt" file with another word (provide the original word and the word to replace it with).

```
cdac@NSL-LW-10001728: ~/Li × + v
cdac@NSL-LW-10001728:~/unzipfile/docs$ cd ..
cdac@NSL-LW-10001728:~/unzipfile$ cd .
cdac@NSL-LW-10001728:~/unzipfile$ cd /
cdac@NSL-LW-10001728:/$ ls
bin          boot  etc   init  lib.usr-is-merged  lost+found  mnt  proc  run  sbin.usr-i
bin.usr-is-merged  dev   home  lib   lib64              media       opt  root  sbin  snap
cdac@NSL-LW-10001728:/$ cd /home/cdac/
cdac@NSL-LW-10001728:~$ ls
Day3  LinuxAssignments  shell.sh  shell3.sh  shell5.sh  shell7.sh  shell9.sh  unzipfile
Lab   loop.sh           shell2.sh shell4.sh  shell6.sh  shell8.sh  switchcase.sh
cdac@NSL-LW-10001728:~$ cd LinuxAssignments/
cdac@NSL-LW-10001728:~/LinuxAssignments$ ls
docs  docs.zip  file1.txt  file3.txt
cdac@NSL-LW-10001728:~/LinuxAssignments$ vi file1.txt
cdac@NSL-LW-10001728:~/LinuxAssignments$ cat file1.txt
Hey, Linux!!

Good Morning Linux. I was asked to add a new text using a text editor so I am doing it. Hehe

Assignments you know. Can't disappoint Adi Sir now.
cdac@NSL-LW-10001728:~/LinuxAssignments$ |
```

Problem 2: Read the instructions carefully and answer accordingly. If there is any need to insert some data then do that as well.

a. Suppose you have a file named "data.txt" containing important information. Display the first 10 lines of this file to quickly glance at its contents using a command.

```
cdac@NSL-LW-10001728:~/LinuxAssignments$ head -10 data.txt
Introduction: In today's fast-paced world, consistency and clarity have become more important than ever. As organizations and individuals strive to communicate effectively, the need for structured and coherent information grows. Communication is the backbone of collaboration, and without a clear message, confusion can easily arise. Every message should be crafted with care to ensure that the intended audience understands it fully and accurately.

Building on this foundation, it is essential to remember that clarity does not mean simplicity alone, but the right balance of detail and conciseness. While too much information can overwhelm the reader, too little can leave important questions unanswered. This balance is difficult to achieve but crucial for successful communication. Consistency in language, tone, and style further strengthens the message, helping to build trust and reliability over time.

Furthermore, when information is presented consistently, it allows for easier processing and retention. Readers are more likely to grasp key points and act upon them when the structure follows a logical pattern. Effective communication also requires understanding the audience's needs, preferences, and background to tailor the content accordingly. Without this awareness, messages may fall flat or even cause misunderstandings.

Moreover, in professional settings, consistent communication practices streamline workflows and reduce errors. Teams working towards common goals benefit from a shared understanding of terminology and expectations. This shared clarity fosters collaboration and encourages innovation. On the other hand, inconsistent or unclear messages can create friction and delay progress.

In addition, technological advancements have changed the way people communicate, making it both easier and more challenging to maintain consistency. Digital platforms allow for rapid information exchange but also increase the risk of misinterpretation if messages are not carefully crafted. It is therefore imperative to adapt communication strategies to fit the medium while preserving clarity.
```

b. Now, to check the end of the file for any recent additions, display the last 5 lines of "data.txt" using another command.

```
cdac@NSL-LW-10001728:~/LinuxAssignments$ ls
data.txt docs docs.zip file1.txt file3.txt
cdac@NSL-LW-10001728:~/LinuxAssignments$ tail -5 data.txt
Finally, ongoing evaluation and feedback are essential components of maintaining effective communication. By regularly reviewing how messages are received and understood, improvements can be made to enhance clarity and consistency further. This iterative process ensures that communication remains a powerful tool rather than a source of confusion.

In summary, clear and consistent communication is foundational for success in any field. It requires deliberate effort, attention to detail, and an understanding of the audience. By prioritizing these elements, individuals and organizations can foster stronger relationships, achieve better outcomes, and navigate the complexities of modern interaction with confidence.
```

c. In a file named "numbers.txt," there are a series of numbers. Display the first 15 lines of this file to analyze the initial data set.

```
cdac@NSL-LW-10001728:~/LinuxAssignments$ head -15 numbers.txt
1
2
3
4
5
6
7
8
8
8
9
10
01
11
12
cdac@NSL-LW-10001728:~/LinuxAssignments$ |
```


d. To focus on the last few numbers of the dataset, display the last 3 lines of "numbers.txt".

```
cdac@NSL-LW-10001728:~/LinuxAssignments$ tail -3 numbers.txt
28
29
30
```

e. Imagine you have a file named "input.txt" with text content. Use a command to translate all lowercase letters to uppercase in "input.txt" and save the modified text in a new file named "output.txt."

```
cdac@NSL-LW-10001728:~/LinuxAssignments$ tr '[:lower:]' '[:upper:]' < input.txt > output.txt
cdac@NSL-LW-10001728:~/LinuxAssignments$ cat output.txt
INTRODUCTION: IN TODAY'S FAST-PACED WORLD, CONSISTENCY AND CLARITY HAVE BECOME MORE IMPORTANT THAN EVER. AS ORGANIZATIONS AND INDIVIDUALS STRIVE TO COMMUNICATE EFFECTIVELY, THE NEED FOR STRUCTURED AND COHERENT INFORMATION GROWS. COMMUNICATION IS THE BACKBONE OF COLLABORATION, AND WITHOUT A CLEAR MESSAGE, CONFUSION CAN EASILY ARISE. EVERY MESSAGE SHOULD BE CRAFTED WITH CARE TO ENSURE THAT THE INTENDED AUDIENCE UNDERSTANDS IT FULLY AND ACCURATELY.

BUILDING ON THIS FOUNDATION, IT IS ESSENTIAL TO REMEMBER THAT CLARITY DOES NOT MEAN SIMPLICITY ALONE, BUT THE RIGHT BALANCE OF DETAIL AND CONCISENESS. WHILE TOO MUCH INFORMATION CAN OVERWHELM THE READER, TOO LITTLE CAN LEAVE IMPORTANT QUESTIONS UNANSWERED. THIS BALANCE IS DIFFICULT TO ACHIEVE BUT CRUCIAL FOR SUCCESSFUL COMMUNICATION. CONSISTENCY IN LANGUAGE, TONE, AND STYLE FURTHER STRENGTHENS THE MESSAGE, HELPING TO BUILD TRUST AND RELIABILITY OVER TIME.

FURTHERMORE, WHEN INFORMATION IS PRESENTED CONSISTENTLY, IT ALLOWS FOR EASIER PROCESSING AND RETENTION. READERS ARE MORE LIKELY TO GRASP KEY POINTS AND ACT UPON THEM WHEN THE STRUCTURE FOLLOWS A LOGICAL PATTERN. EFFECTIVE COMMUNICATION ALSO REQUIRES UNDERSTANDING THE AUDIENCE'S NEEDS, PREFERENCES, AND BACKGROUND TO TAILOR THE CONTENT ACCORDINGLY. WITHOUT THIS AWARENESS, MESSAGES MAY FALL FLAT OR EVEN CAUSE MISUNDERSTANDINGS.

MOREOVER, IN PROFESSIONAL SETTINGS, CONSISTENT COMMUNICATION PRACTICES STREAMLINE WORKFLOWS AND REDUCE ERRORS. TEAMS WORKING TOWARDS COMMON GOALS BENEFIT FROM A SHARED UNDERSTANDING OF TERMINOLOGY AND EXPECTATIONS. THIS SHARED CLARITY FOSTERS COLLABORATION AND ENCOURAGES INNOVATION. ON THE OTHER HAND, INCONSISTENT OR UNCLEAR MESSAGES CAN CREATE FRICTION AND DELAY PROGRESS.

IN ADDITION, TECHNOLOGICAL ADVANCEMENTS HAVE CHANGED THE WAY PEOPLE COMMUNICATE, MAKING IT BOTH EASIER AND MORE CHALLENGING TO MAINTAIN CONSISTENCY. DIGITAL PLATFORMS ALLOW FOR RAPID INFORMATION EXCHANGE BUT ALSO INCREASE THE RISK OF MISINTERPRETATION IF MESSAGES ARE NOT CAREFULLY CRAFTED. IT IS THEREFORE IMPERATIVE TO ADAPT COMMUNICATION STRATEGIES TO FIT THE MEDIUM WHILE PRESERVING CLARITY.

FINALLY, ONGOING EVALUATION AND FEEDBACK ARE ESSENTIAL COMPONENTS OF MAINTAINING EFFECTIVE COMMUNICATION. BY REGULARLY REVIEWING HOW MESSAGES ARE RECEIVED AND UNDERSTOOD, IMPROVEMENTS CAN BE MADE TO ENHANCE CLARITY AND CONSISTENCY FURTHER. THIS ITERATIVE PROCESS ENSURES THAT COMMUNICATION REMAINS A POWERFUL TOOL RATHER THAN A SOURCE OF CONFUSION.

IN SUMMARY, CLEAR AND CONSISTENT COMMUNICATION IS FOUNDATIONAL FOR SUCCESS IN ANY FIELD. IT REQUIRES DELIBERATE EFFORT, ATTENTION TO DETAIL, AND AN UNDERSTANDING OF THE AUDIENCE. BY PRIORITIZING THESE ELEMENTS, INDIVIDUALS AND ORGANIZATIONS CAN FOSTER STRONGER RELATIONSHIPS, ACHIEVE BETTER OUTCOMES, AND NAVIGATE THE COMPLEXITIES OF MODERN INTERACTION WITH CONFIDENCE.
```

f. In a file named "duplicate.txt," there are several lines of text, some of which are duplicates. Use a command to display only the unique lines from "duplicate.txt."

```
cdac@NSL-LW-10001728:~/LinuxAssignments$ cat > duplicate.txt
We often say that practice makes perfect, and perfect practice makes excellence.
Failure is not the end; it is simply a step, a step toward growth and learning.
We often say that practice makes perfect, and perfect practice makes excellence.
It's important to believe, to believe in ourselves even when others don't.
cdac@NSL-LW-10001728:~/LinuxAssignments$ cat duplicate.txt
We often say that practice makes perfect, and perfect practice makes excellence.
Failure is not the end; it is simply a step, a step toward growth and learning.
We often say that practice makes perfect, and perfect practice makes excellence.
It's important to believe, to believe in ourselves even when others don't.
cdac@NSL-LW-10001728:~/LinuxAssignments$ sort duplicate.txt | uniq
Failure is not the end; it is simply a step, a step toward growth and learning.
It's important to believe, to believe in ourselves even when others don't.
We often say that practice makes perfect, and perfect practice makes excellence.
cdac@NSL-LW-10001728:~/LinuxAssignments$
```

g. In a file named "fruit.txt," there is a list of fruits, but some fruits are repeated. Use a command to display each unique fruit along with the count of its occurrences in "fruit.txt."

```
Watermelon
Cherries
cdac@NSL-LW-10001728:~/LinuxAssignments$ cat fruit.txt
Banana
Grapes
Apple
Apple
Orange
Pineapple
Banana
Watermelon
Cherries
cdac@NSL-LW-10001728:~/LinuxAssignments$ uniq fruit.txt
Banana
Grapes
Apple
Orange
Pineapple
Banana
Watermelon
Cherries
cdac@NSL-LW-10001728:~/LinuxAssignments$ sort fruit.txt | uniq
Apple
Banana
Cherries
Grapes
Orange
Pineapple
Watermelon
cdac@NSL-LW-10001728:~/LinuxAssignments$ |
```